

Design and Implementation of a Custom Reliable Transport Protocol over UDP

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Abstract

The Transmission Control Protocol (TCP) provides reliable, ordered, congestion-aware delivery over an unreliable network layer, but its internal mechanisms are often studied only theoretically, leaving a gap between conceptual understanding and hands-on implementation experience. This project bridges that gap by designing and implementing RUDP (Reliable UDP), a custom transport-layer protocol built entirely on UDP sockets that replicates and experiments with the core reliability and congestion-control mechanisms found in TCP, while remaining flexible enough to explore design alternatives.

The protocol implements connection establishment via a simplified three-way handshake, reliable in-order delivery using sequence numbers and cumulative acknowledgments, retransmission on timeout and triple-duplicate ACKs (fast retransmit), sliding-window flow control, and AIMD-based congestion control with slow start and congestion avoidance phases, alongside a Cubic-inspired variant for comparative analysis. Correctness and robustness are validated against packet loss, reordering, and duplication scenarios. The implementation is tested under emulated network conditions, including variable latency, jitter, and packet loss, using Linux traffic control (tc/netem), and benchmarked against native TCP and UDP across throughput, round-trip latency, retransmission rate, and fairness under contention.

Deliverables include a fully functional client-server implementation, a quantitative performance evaluation comparing congestion-control strategies under varying network conditions, and visualizations of congestion window evolution, throughput trends, and loss-recovery behavior over time, culminating in a written analysis of design trade-offs and observed protocol dynamics.